

Processes, Ports, Services, and Real System Understanding

These notes were written by Kevin Beideman. The recommended amount of time to spend on these topics for adequate knowledge and understanding of the topic is 3 hours and 30 minutes.

Goal: Start understanding how a REAL operating system behaves internally.

Processes + Task Manager Internals

Video: [What is a Process in an Operating System?](#)

- What is a Process?

Article: [What Is a Port? Network Ports & Port Numbers Explained](#)

What is a Process? A program that is currently running in memory. The OS treats it as a unit of work. For example, opening Google Chrome -> creates Chrome processes. More examples are Discord.exe and Spotify.exe.

What is a Service? A service is a background process that performs system or application tasks automatically. An example is Windows Update or Antivirus service.

What is a Port? A port is a logical communication endpoint used by networking applications. Ports allow multiple network services to exist on one computer at the same time. Ports range from 0 to 65535 and work alongside IP addresses, which then pinpoint a specific device on the network. Example: if something arrives on port 21, the server knows to forward it to the File Transfer Protocol (FTP) server.

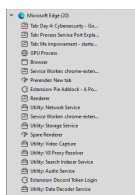
Active Connections:

Investigation Notes:

Now, we will take a Deep Dive within the Task Manager. Open it.

Task: Find: Chrome, Discord, Steam. Write: memory usage, CPU usage, and process names.

I use Edge personally. 1,530 MB of memory usage, 5.5% CPU, process names I will screenshot. (Recommend looking at them in the task manager, hard to see).



Discord: 372.0 MB of memory usage, 0.1% CPU, Discord.exe

.... And so on...

Now, go to the Details Tab. You'll see PID, status, usernames, and process names.

PID = Process ID. Every running process gets a unique identifier.

We will now look at Command Line Process Analysis. Open Powershell.

Run *tasklist*. This lists running processes.

Now try: *tasklist | findstr chrome*

[*tasklist*](#) | [Microsoft Learn](#)

```
PS C:\Windows\system32> tasklist | findstr Discord
Discord.exe           14876 Console           1      67,628 K
Discord.exe           15016 Console           1       2,660 K
Discord.exe           15204 Console           1     37,044 K
Discord.exe           15248 Console           1     21,308 K
Discord.exe            6432 Console           1    478,748 K
Discord.exe           15812 Console           1       8,492 K
```

This lets you filter for specific processes.

Now, let's kill a process. Open Notepad. Then run, *tasklist | findstr notepad*.

```
PS C:\Users\1leaf> tasklist | findstr notepad
notepad.exe           22088 Console           1     14,672 K
```

Find the PID from the command. Then run: *taskkill /PID (PID HERE) /F*

This will close Notepad!

Ports + Network Connections

What is a port? A port = a communication endpoint. The easiest way to think about this is: IP address = apartment building, and the port = apartment number.

There are important ports. View the table below.

Port	Purpose
80	HTTP
443	HTTPS
22	SSH
53	DNS
3389	Remote Desktop

Just a few examples...

Article: [What is a computer port? | Ports in networking](#)

Let's do some live work. Open powershell and run the command **netstat -ano**. This will let us view active connections.

You should see some columns. The protocol refers to TCP or UDP. The local address is your machine, and the foreign address is the remote system. The state is ESTABLISHED or LISTENING, and the PID is the owning process.

Find an **ESTABLISHED** connection. Note the **PID**. Now run: **tasklist | findstr PIDNUMBER**

```
PS C:\Users\lleaf> tasklist | findstr 21968
EpicGamesLauncher.exe      21968 Console          1      51,768 K
```

You can also filter the results. For example, lets only list **LISTENING** connections. Try: **netstat -ano | findstr LISTENING**.

Next, try: **netstat -ano | findstr :443**.

Services + Background Services

A service = a background software that runs continuously. (Windows Update)

To open Services, Press: **Windows + R** and type **services.msc**.

Task: Find **Windows Update**, **DHCP Client**, **DNS Client**, and **Printer Spooler**. Write **status** and **startup type**.

Windows Update: Status = Running and Startup Type = Manual
DHCP Client: Running and Automatic
DNS Client: Running and Automatic
Printer Spooler: Running and Automatic

Let's look at the Command Line Services again. Open Powershell and run **Get-Service**. This will list all the services. You can filter like Ports and Processes. Try **Get-Service | findstr Running**. Next, try **Get-Service | findstr DNS**.

```
PS C:\Windows\system32> Get-Service | findstr DNS
Running Dnscache           DNS Client
```

Now, let's Stop a Service. Find **Print Spooler** and stop it manually in Services. Then start it again.

Mini Investigation Project!

A user has told you a computer feels slow and suspicious. We must now investigate the machine.

Investigate:

Processes:

- Highest CPU? Microsoft Edge
- Highest RAM? Microsoft Edge.

Connections:

- Active Established Connections? **Netstat -ano | findstr ESTABLISHED**
- Listening ports? **Netstat -ano | findstr LISTENING**

Check services that might be suspicious or unnecessary.